Exercise J01

Q1. The most general solution of $\tan\theta = -1$, $\cos\theta = \dfrac{1}{\sqrt{2}}$ is :

(A) $n\pi + \dfrac{7\pi}{4}$ (B) $n\pi + (-1)^n\dfrac{7\pi}{4}$ (C) $2n\pi + \dfrac{7\pi}{4}$ (D) None of these

Ans. C

Sol. -

Q2. $\dfrac{\sin 3\theta}{2\cos 2\theta + 1} = \dfrac{1}{2}$, if :

(A) $\theta = n\pi + \dfrac{\pi}{6}$ (B) $\theta = 2n\pi - \dfrac{\pi}{6}$ (C) $\theta = n\pi + (-1)^n\dfrac{\pi}{6}$ (D) $\theta = n\pi - \dfrac{\pi}{6}$

Ans. C

Sol. -

Q3. Find no. of roots of the equation tan x + sec x = 2 cos x in the interval $[0, 2\pi]$-

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. -

Q4. The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2\sin^2 x + 5\sin x - 3 = 0$ is –

(A) 6 (B) 1 (C) 2 (D) 4

Ans. D

Sol. -

Q5. If $0 < x < \pi$, and $\cos x + \sin x = \dfrac{1}{2}$, then tan x is –

(A) $(4 - \sqrt{7})/3$ (B) $-(4 + \sqrt{7})/3$ (C) $(1 + \sqrt{7})/4$ (D) $(1 - \sqrt{7})/4$

Ans. B

Sol. $0 < x < \pi$ and $\cos x + \sin x = \dfrac{1}{2}$

$2(\cos x + \sin x) = 1$

$\Rightarrow 4[\cos^2 x + \sin^2 x + 2\sin x\cos x] = 1$

$\Rightarrow 4[1 + \tan^2 x + 2\tan x] = \sec^2 x$

$\Rightarrow 4 + [1 + \tan^2 x + 2\tan x] = 1 + \tan^2 x$

$\Rightarrow 4 + 4\tan^2 x + 8\tan x = 1 + \tan^2 x$

$\Rightarrow 3\tan^2 x + 8\tan x + 1 = 0$

$\tan x = \dfrac{-8 \pm \sqrt{64 - 4.3.3}}{2.3} = \dfrac{-8 \pm 2\sqrt{2}}{2.3} = \dfrac{-4 \pm \sqrt{7}}{3}$

Q6. Let P = $\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta)$ and Q = $\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta)$ be two sets. Then

(A) $P \subset Q$ and $Q - P \neq \phi$ (B) $Q \subset P$ (C) $P \subset Q$ (D) $P = Q$

Ans. D

Sol. -

Q7. The general solution of, sin x - 3 sin 2x + sin 3x = cos x - 3 cos 2x + cos 3x is

(A) $n\pi + \dfrac{\pi}{8}$ (B) $\dfrac{n\pi}{2} + \dfrac{\pi}{8}$ (C) $(-1)^n\left(\dfrac{n\pi}{2}\right) + \dfrac{\pi}{8}$ (D) $2n\pi + \cos^{-1}\left(\dfrac{3}{2}\right)$

Ans. B

Sol. -

Q8. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are.

(A) $\dfrac{2\pi}{9}, \dfrac{\pi}{4}, \dfrac{4\pi}{9}, \dfrac{\pi}{2}, \dfrac{3\pi}{4}, \dfrac{8\pi}{9}$ (B) $\dfrac{\pi}{4}, \dfrac{5\pi}{12}, \dfrac{\pi}{2}, \dfrac{2\pi}{3}, \dfrac{3\pi}{4}, \dfrac{8\pi}{9}$ (C) $\dfrac{2\pi}{9}, \dfrac{\pi}{4}, \dfrac{\pi}{2}, \dfrac{2\pi}{3}, \dfrac{3\pi}{4}, \dfrac{35\pi}{36}$

(D) $\dfrac{2\pi}{9}, \dfrac{\pi}{4}, \dfrac{\pi}{2}, \dfrac{2\pi}{3}, \dfrac{3\pi}{4}, \dfrac{8\pi}{9}$

Ans. A

Sol. -

Q9. If $0 \leq x < 2p$, then the number of real values of x, which satisfy the equation
cosx + cos2x + cos3x + cos4x = 0, is :

(A) 7 (B) 9 (C) 3 (D) 5

Ans. A

Sol. -

Q10. If sum of all the solution of the equation $8\cos\left(\cos\left(\dfrac{\pi}{6} + x\right). \cos\left(\dfrac{\pi}{6} - x\right)1 - \dfrac{1}{2}\right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to

(A) $\dfrac{8}{9}$ (B) $\dfrac{20}{9}$ (C) $\dfrac{2}{3}$ (D) $\dfrac{13}{9}$

Ans. D

Sol. -

Q11. If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of cos 4x is

(A) $\frac{2}{9}$ (B) $-\frac{7}{9}$ (C) $-\frac{3}{5}$ (D) $\frac{1}{3}$

Ans. B

Sol. -

Q12. If $\theta \in [0, 5\pi]$ and $r \in R$ such that $2\sin\theta = r^4 - 2r^2 + 3$ then the maximum number of values of the pair (r, θ) is

(A) 8 (B) 10 (C) 6 (D) none of these

Ans. C

Sol. -

Q13. The number of real solutions of , $\sin e^x . \cos e^x = 2^{x-2} + 2^{-x-2}$ is :

(A) Zero (B) One (C) Two (D) Infinite

Ans. A

Sol. -

Q14. The general solution of the equation , $\cos x \ \cos 6x = -1$ is :

(A) $x = (2n+1)\pi$, $n \in I$ (B) $x = 2n\pi$, $n \in I$ (C) $x = (4n-1)\pi$, $n \in I$

(D) None of these

Ans. A

Sol. -

Q15. The number of solutions of the equation , $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$ is :

(A) Zero (B) Only one (C) Only two (D) Greater than 2

Ans. B

Sol. -

Q16. General solution for θ if , $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$ is :

(A) $2n\pi + \frac{7\pi}{6}$ (B) $2n\pi + \frac{\pi}{6}$ (C) $2n\pi - \frac{7\pi}{6}$ (D) None of these

Ans. A

Sol. -

Q17. The number of values of ' x ' for which $\sin 2x + \cos 4x = 2$ is :

(A) 0 (B) 1 (C) 2 (D) infinite

Ans. A

Sol. $\sin 2x + \sin 4x = 2$

$\sin 2x = 1, \sin 4x =$ `

$2x = n\pi + (-1)^n \frac{\pi}{2}, 4x = m\pi + (-1)^m \frac{\pi}{2}$

$x = \frac{n\pi}{2} (-1)^n \frac{\pi}{4}, x = \frac{m\pi}{4}, x = \frac{m\pi}{4} + (-1)^m \frac{\pi}{8}$

No Common solution

Q18. The equation , $4 \sin^2 x + 4 \sin x + a^2 - 3 = 0$ possesses a solution if 'a' belongs to the interval :

(A) $(-1 , 3)$ (B) $(-3 , 1)$ (C) $[-2 , 2]$ (D) $R - (-2 , 2)$

Ans. C

Sol. -

Q19. The equation, $\cos 2x + a \sin x = 2a - 7$ possesses a solution if :

(A) $a < 2$ (B) $2 \leq a \leq -6$ (C) $a > 6$ (D) a is any integer

Ans. B

Sol. $\cos 2x + a \sin x = 2a - 7$

i.e. $2\sin^2 x - \sin x + 2a - 8 = 0$

$\sin x = \frac{a \pm \sqrt{a^2 - 8(2a - 8)}}{4} = \frac{a \pm (a - 8)}{4}$

$\sin x = \frac{a - 4}{2}$ or 2

Hence $-1 \leq (a - 4)/2 \leq 1 \Rightarrow$ the range of a

Q20. If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is

(A) $\left(0, \frac{\pi}{3}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$ (C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48}, \pi\right)$

Ans. A

Sol. -

Q21. $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = 1/e$, where $\alpha, \beta \in [-\pi, \pi]$, numbers of pairs of α, β which satisfy both the equations is

(A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. -

Q22. The number of solutions of the pair of equations $2\sin^2\theta - \cos 2\theta = 0$ $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $2\sin^2\theta - \cos 2\theta = 0$

$\Rightarrow \sin^2\theta = \frac{1}{4}$

Also, $2\cos^2\theta – 3\sin\theta = 0$

$\Rightarrow \sin\theta = \frac{1}{2}$

$\Rightarrow$ two solutions exist in the interval $[0, 2\pi]$